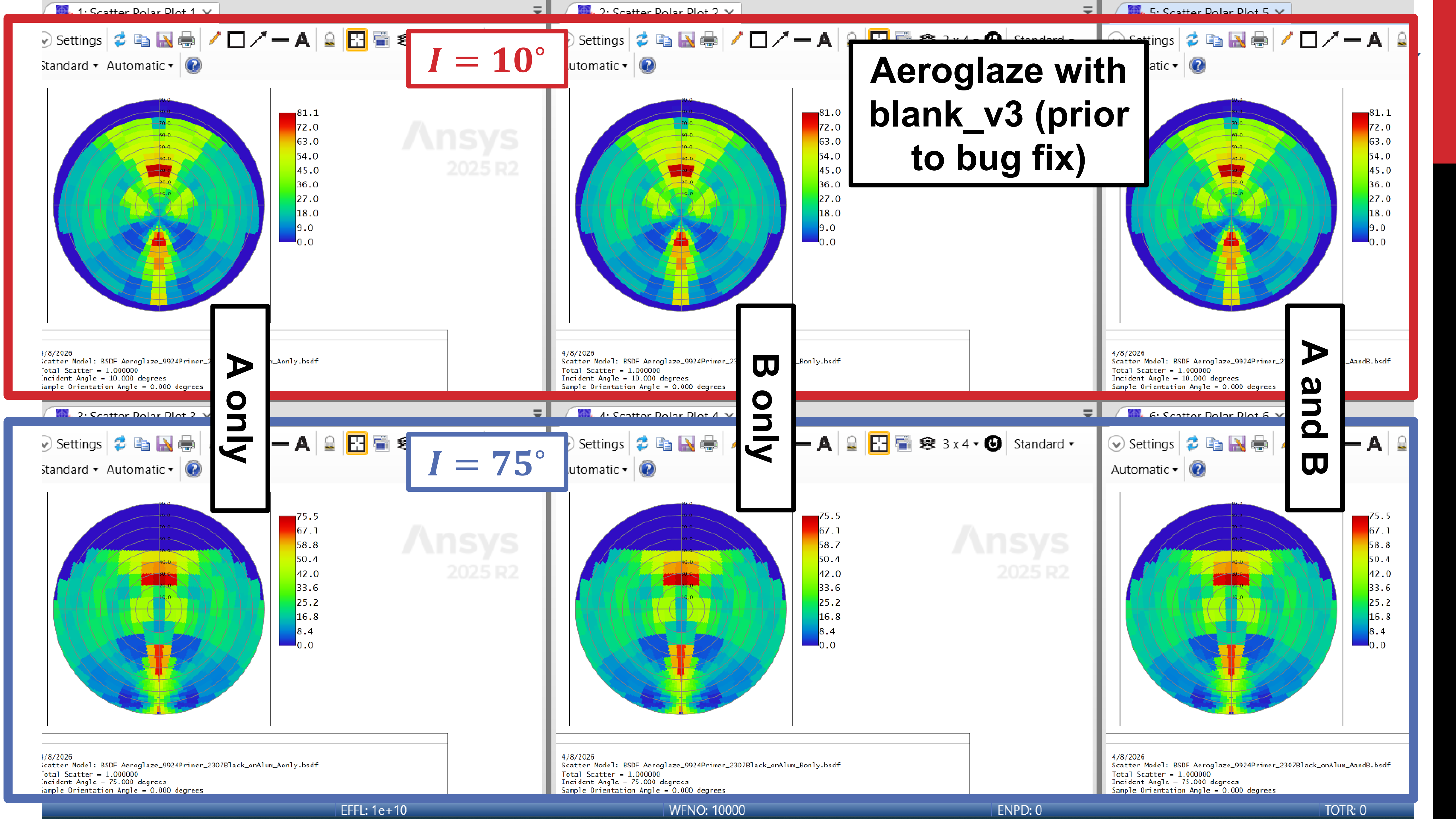
Aeroglaze measurement

2026/04/08

Discussion on measurement result

- Initially, surface seems scratched, otherwise bug in code (but unexpected since code shown to work for NiTE samples and IMX455)
- Or, might just be property of surface that “specular” beam splits into the two “upper” and “lower” regions seen in polar plots

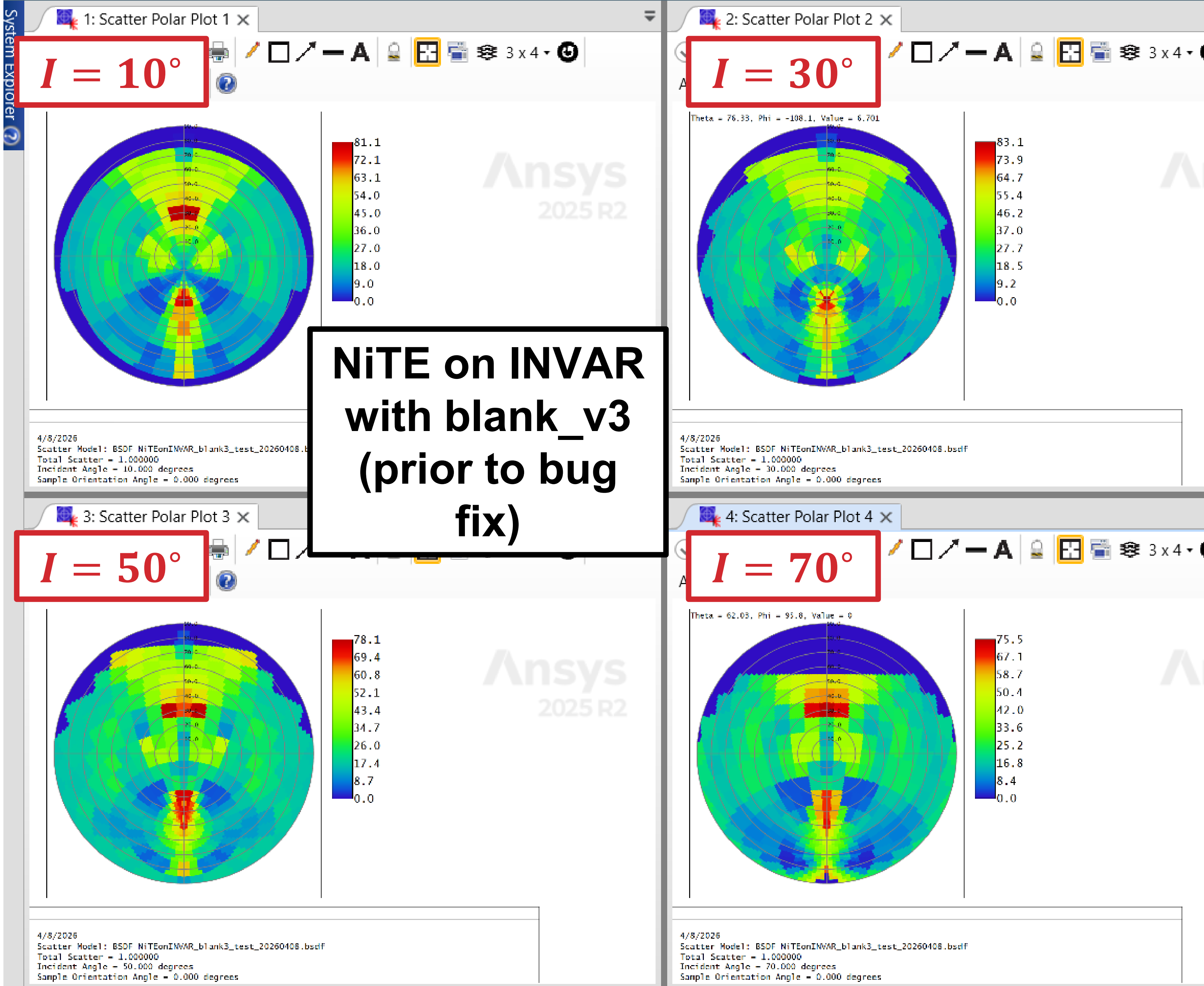
- **CONCLUSION:** the new blank reference “blank_v3o0_20260407.xls” is what is off
- Also, there was bug in how sample and blank datasets without identical angles were reduced to their overlapping angles… but this appears not to fix the initial observation hence why blank reference was concluded to be the source of error

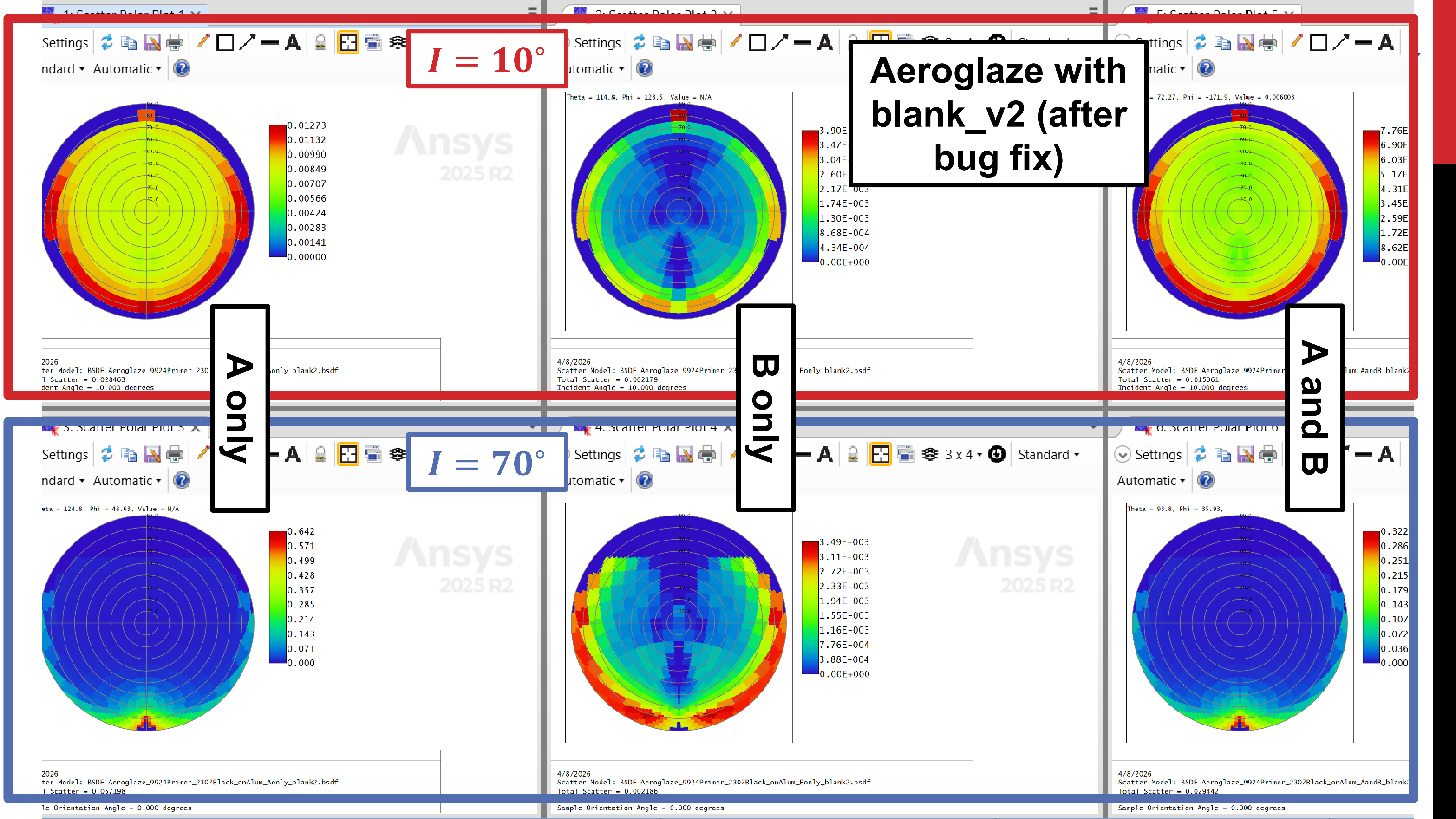


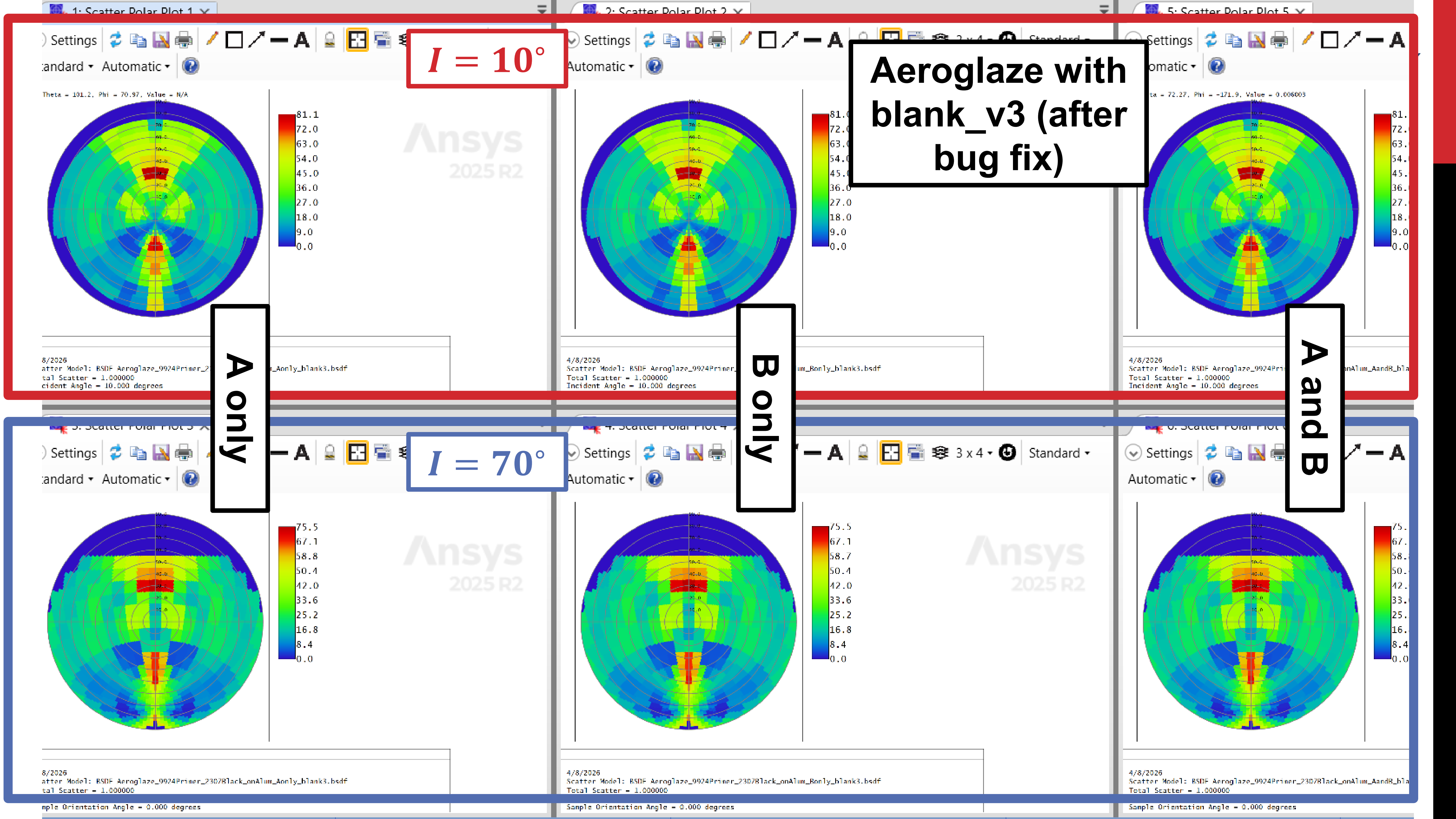
I = 10°
Aeroglaze with blank_v3 (prior to bug fix)
A only
B only
A and B
I = 75°
EFFL: 1e+10
WFNO: 10000
ENPD: 0
TOTR: 0

- NiTE on INVAR shows same pattern
  - Conclusion:
    this is result of "blank_v3o0_20260407.xls" data, not of sample data

- So, reprocessing Aeroglaze with blank v2o0 data




$I = 10°$
Aeroglaze with blank_v2 (after bug fix)
A only
B only
A and B
$I = 70°$


Aeroglaze with blank_v3 (after bug fix)
$I = 10°$
$I = 70°$
A only
B only
A and B

Discussion

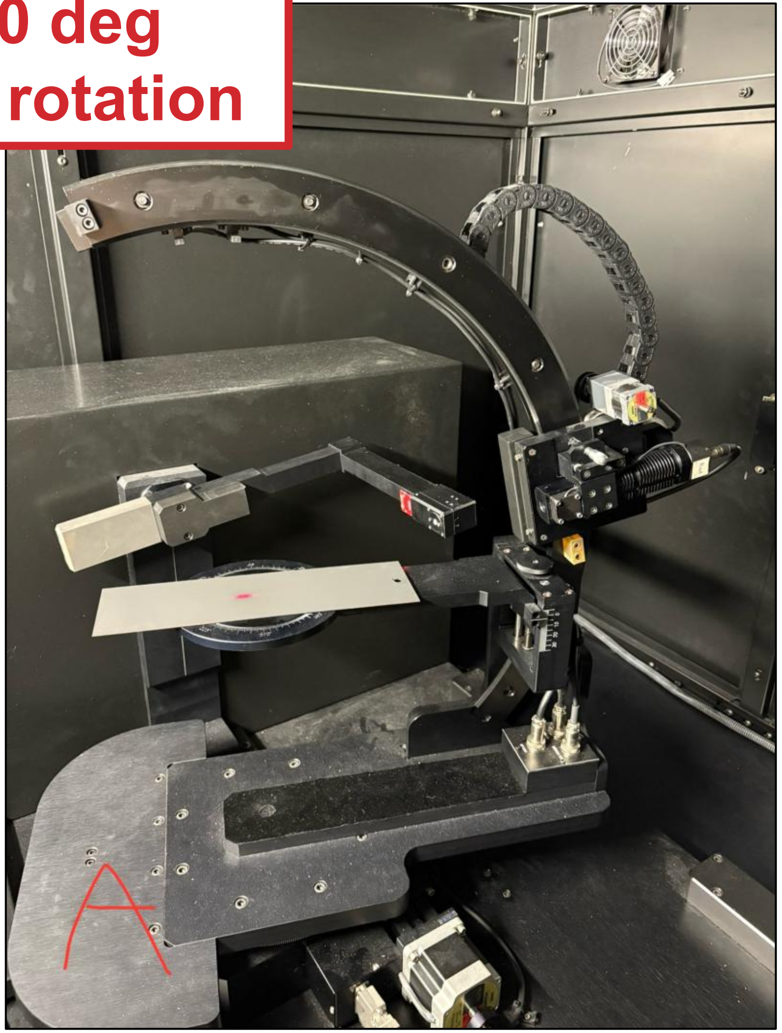
- The "bug" was that a new method in data processing (in 'process_rt300s.m' function) was implemented to allow for sample and blank measurement files to not have identical angles, by masking-out non-overlapping angles
  - However, this assumed the blank angles "encompassed" the sample angles, so only the extra blank angles were filtered out
  - The "bug fix" was doing this both ways… so it was thought maybe this was why the "blank_v3o0_20260407.xls" file produced weird polar plots
    - But, even after this fix, the plots look similar … so most likely it is this blank file itself that is inaccurate

Conclusion

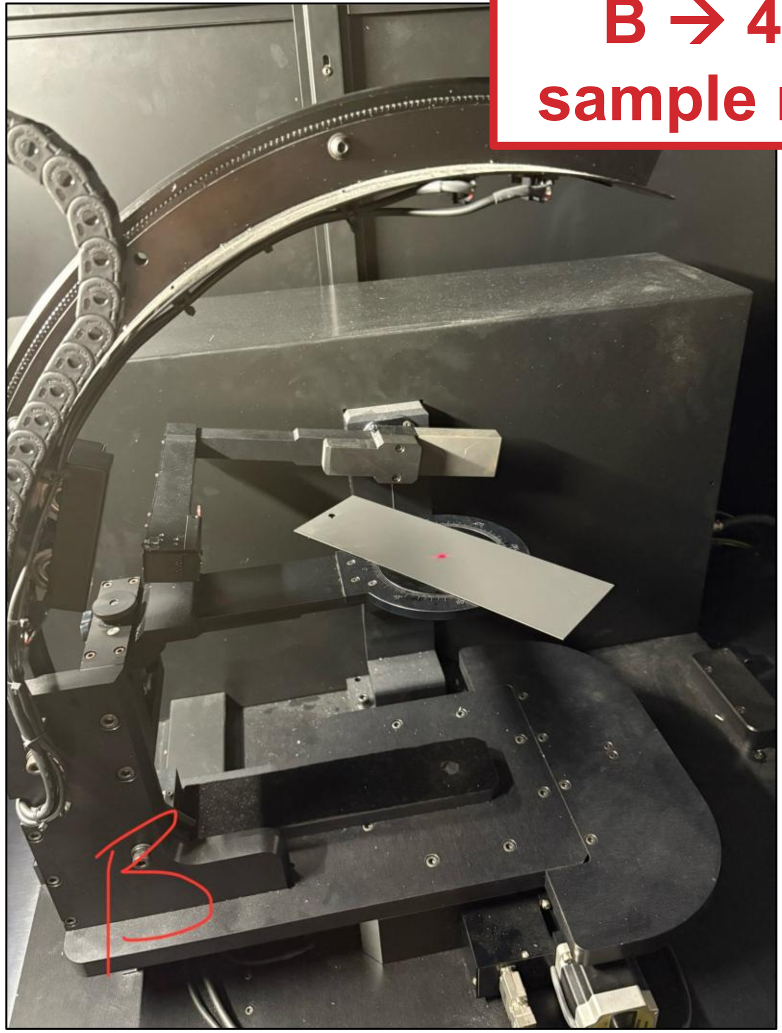
- Since “blank_v2o0_20250911.xls” is already trusted to behave well, using that for Aeroglaze (even though it is lower angular resolution)
- But… should A, B, or (A and B) be used?
  - ‘A’ is sample rotation of 0 deg
  - ‘B’ is sample rotation of 45 deg
  - The thought was that the sample was isotropic and therefore these two measurements could be averaged to give a more accurate overall result

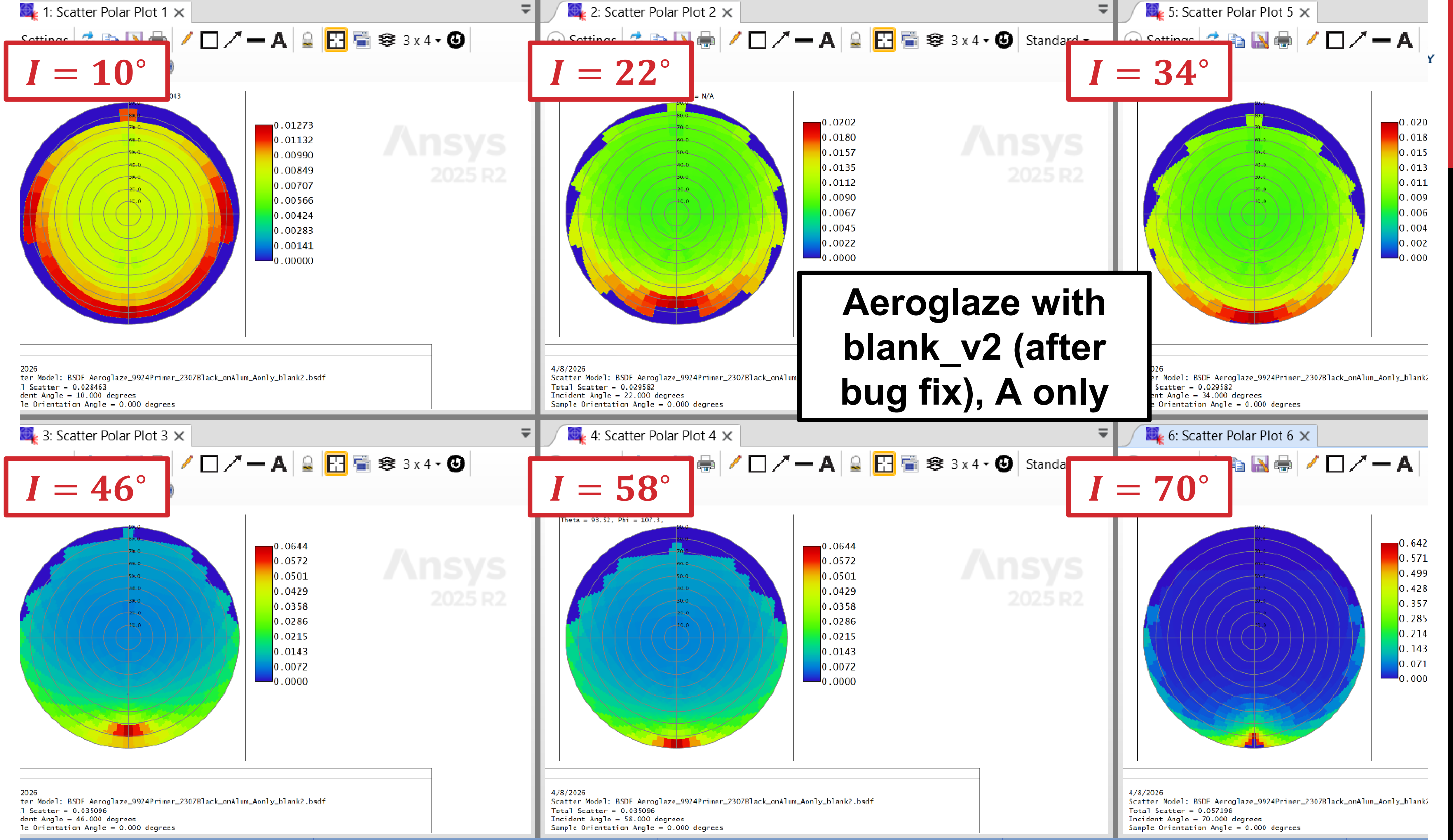
Measurement setup

A → 0 deg
sample rotation

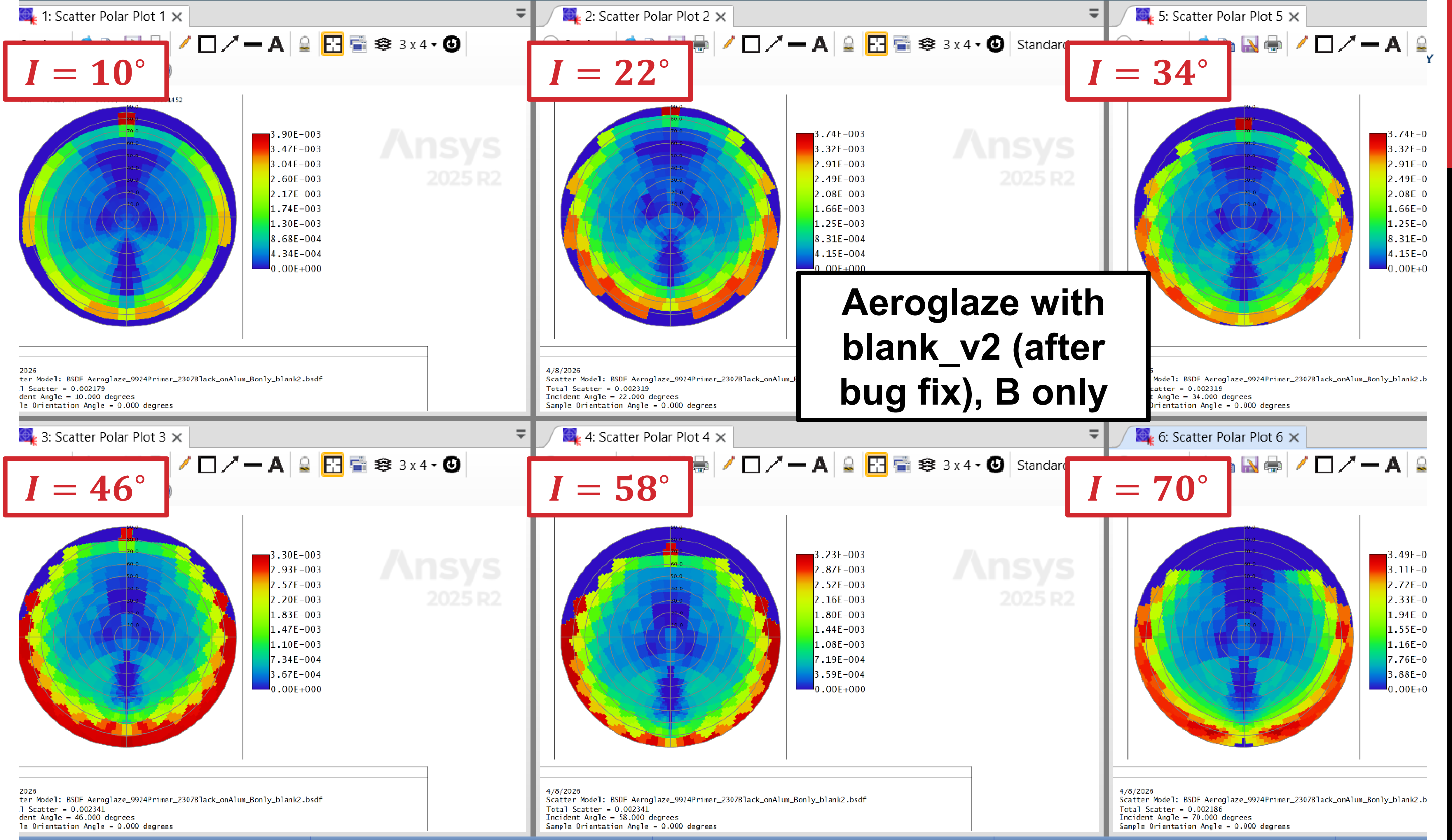


B → 45 deg
sample rotation

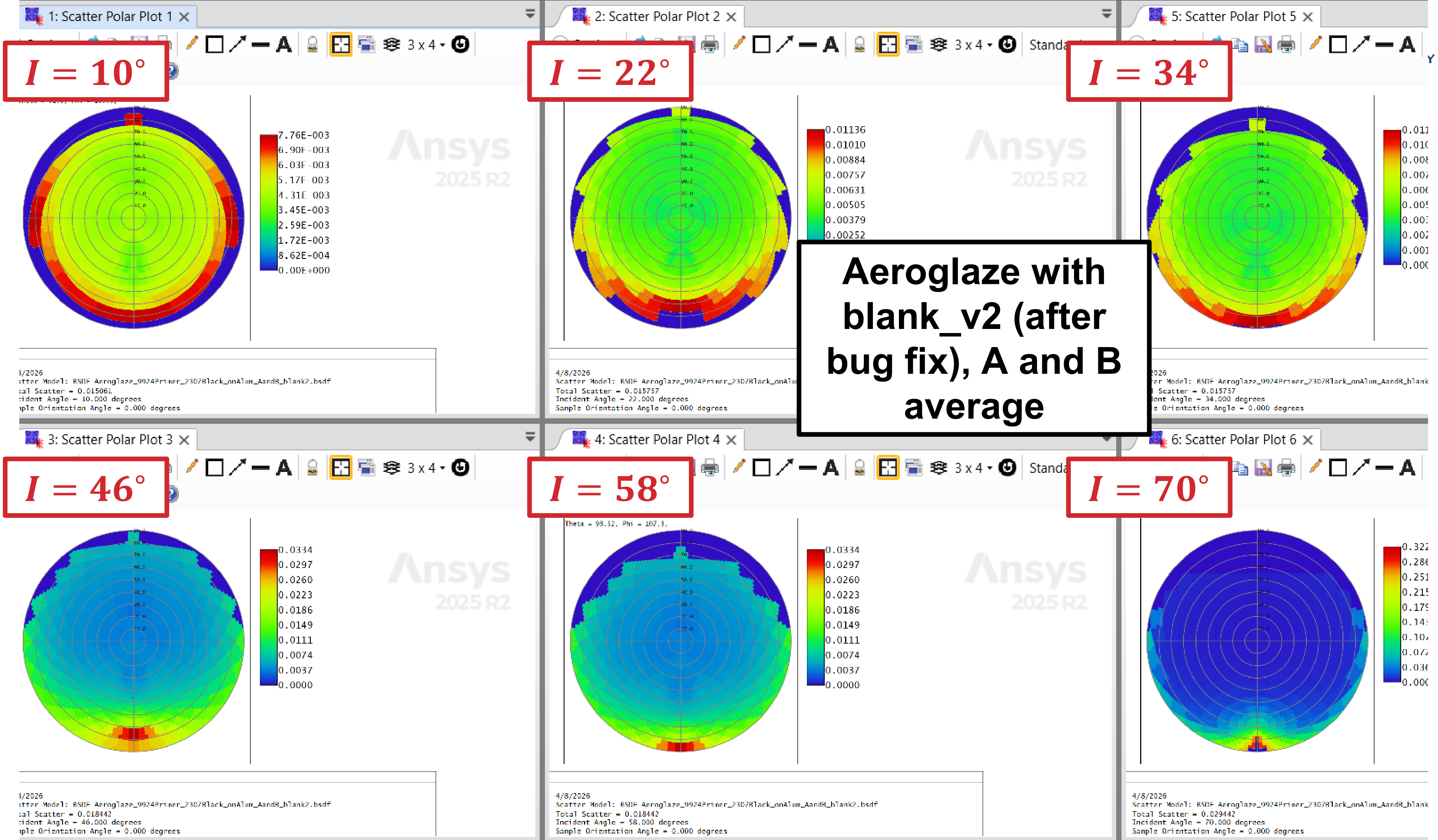




Aeroglaze with blank_v2 (after bug fix), A only



Aeroglaze with blank_v2 (after bug fix), B only
1: Scatter Polar Plot 1
$I = 10°$
3.90E-003
3.47E-003
3.04E-003
2.60E-003
2.17E-003
1.74E-003
1.30E-003
8.68E-004
4.34E-004
0.00E+000
Ansys 2025 R2
2026
ter Model: BSDF Aeroglaze_9924Primer_2307Black_onAlum_Bonly_blank2.bsdf
l Scatter = 0.002179
dent Angle = 10.000 degrees
le Orientation Angle = 0.000 degrees
2: Scatter Polar Plot 2
$I = 22°$
3.74E-003
3.32E-003
2.91E-003
2.49E-003
2.08E-003
1.66E-003
1.25E-003
8.31E-004
4.15E-004
0.00E+000
Ansys 2025 R2
4/8/2026
Scatter Model: BSDF Aeroglaze_9924Primer_2307Black_onAlum_B
Total Scatter = 0.002319
Incident Angle = 22.000 degrees
Sample Orientation Angle = 0.000 degrees
5: Scatter Polar Plot 5
$I = 34°$
3.74E-0
3.32E-0
2.91E-0
2.49E-0
2.08E-0
1.66E-0
1.25E-0
8.31E-0
4.15E-0
0.00E+0
Model: BSDF Aeroglaze_9924Primer_2307Black_onAlum_Bonly_blank2.b
atter = 0.002319
Angle = 34.000 degrees
Orientation Angle = 0.000 degrees
3: Scatter Polar Plot 3
$I = 46°$
3.30E-003
2.93E-003
2.57E-003
2.20E-003
1.83E-003
1.47E-003
1.10E-003
7.34E-004
3.67E-004
0.00E+000
Ansys 2025 R2
2026
ter Model: BSDF Aeroglaze_9924Primer_2307Black_onAlum_Bonly_blank2.bsdf
l Scatter = 0.002341
dent Angle = 46.000 degrees
le Orientation Angle = 0.000 degrees
4: Scatter Polar Plot 4
$I = 58°$
3.23E-003
2.87E-003
2.52E-003
2.16E-003
1.80E-003
1.44E-003
1.08E-003
7.19E-004
3.59E-004
0.00E+000
Ansys 2025 R2
4/8/2026
Scatter Model: BSDF Aeroglaze_9924Primer_2307Black_onAlum_Bonly_blank2.bsdf
Total Scatter = 0.002341
Incident Angle = 58.000 degrees
Sample Orientation Angle = 0.000 degrees
6: Scatter Polar Plot 6
$I = 70°$
3.49E-0
3.11E-0
2.72E-0
2.33E-0
1.94E-0
1.55E-0
1.16E-0
7.76E-0
3.88E-0
0.00E+0
4/8/2026
Scatter Model: BSDF Aeroglaze_9924Primer_2307Black_onAlum_Bonly_blank2.b
Total Scatter = 0.002186
Incident Angle = 70.000 degrees
Sample Orientation Angle = 0.000 degrees


Aeroglaze with blank_v2 (after bug fix), A and B average
1: Scatter Polar Plot 1
I = 10°
7.76E-003
6.90E-003
6.03E-003
5.17E-003
4.31E-003
3.45E-003
2.59E-003
1.72E-003
8.62E-004
0.00E+000
Total Scatter = 0.015061
Incident Angle = 10.000 degrees
2: Scatter Polar Plot 2
I = 22°
0.01136
0.01010
0.00884
0.00757
0.00631
0.00505
0.00379
0.00252
4/8/2026
Total Scatter = 0.015757
Incident Angle = 22.000 degrees
Sample Orientation Angle = 0.000 degrees
5: Scatter Polar Plot 5
I = 34°
Total Scatter = 0.015757
Incident Angle = 34.000 degrees
3: Scatter Polar Plot 3
I = 46°
0.0334
0.0297
0.0260
0.0223
0.0186
0.0149
0.0111
0.0074
0.0037
0.0000
Total Scatter = 0.018442
Incident Angle = 46.000 degrees
4: Scatter Polar Plot 4
I = 58°
Theta = 93.52, Phi = 107.3
0.0334
0.0297
0.0260
0.0223
0.0186
0.0149
0.0111
0.0074
0.0037
0.0000
4/8/2026
Scatter Model: BSDF Aeroglaze_9924Primer_Z307Black_onAlum_AandB_blank2.bsdf
Total Scatter = 0.018442
Incident Angle = 58.000 degrees
Sample Orientation Angle = 0.000 degrees
6: Scatter Polar Plot 6
I = 70°
4/8/2026
Total Scatter = 0.029442
Incident Angle = 70.000 degrees
Sample Orientation Angle = 0.000 degrees
Ansys 2025 R2

Conclusion

- The Aeroglaze sample appears anisotropic
- If assuming isotropic, probably want “A only”
- “B only” looks like incident light is almost fully absorbed, and the red in the plot is driven more so from BRDF = RT/cos(R) → 0 as R → 90 deg, rather than driven by actual scattered light

- **Use:** “Aeroglaze_9924Primer_2307Black_onAlum_Aonly_blank2.bsdf”